Yu Hu

huyuaaaray@gmail.com
(+1) 352-888-9044
https://huyuaaaray.github.io

Healthcare AI researcher with 3+ years of experience developing, evaluating, and deploying machine learning systems in real-world clinical settings. Research interests include **trustworthy clinical AI, model reliability, generalizability, and responsible AI integration into clinical workflows**. Experienced in clinical LLM evaluation, clinician-in-the-loop validation, algorithmic fairness, causal machine learning, and EHR-based predictive modeling. Also developed scalable LLM evaluation frameworks and production-ready MLOps/LLMOps pipelines adopted by three clinical programs.

Education

University of Florida — Gainesville, FL
M.S. in Electrical and Computer Engineering | ***GPA:** 3.56/4.0* — Aug. 2021 – May 2023

Hubei University of Technology — Wuhan, China
B.S. in Electrical Engineering | ***GPA:** 3.3/4.0* — Sep. 2016 – Jun. 2020

Professional Experience

University of Florida — Gainesville, FL

Data Scientist III — Sep. 2025 – Present

- Drive clinical **LLM evaluation** for the Quality & Patient Safety group, comparing models against clinician review and building **LLM-as-a-Judge** workflows that route flagged outputs to clinician review.
- Architected a shared, config-driven **LLMOps** platform adopted by three clinical programs, processing EHR clinical notes with self-hosted long-context inference (**vLLM**, **SLURM**) on the HiPerGator HPC cluster.
- Translate clinical review workflows (readmission, mortality, transitional care) into structured, evidence-grounded LLM pipelines in partnership with clinical teams.
- Develop **machine learning** models on EHR data, including neonatal feeding-risk prediction, from clinician-validated feature definition through on-premise deployment.

Application Programmer — Mar. 2024 – May 2025

- Implemented **MLOps** workflows with CI/CD and monitoring, enabling scalable deployment on **Linux** systems.
- Developed scalable **ETL pipelines** and engineered clinical features using **cluster computing**, boosting ML model performance on large-scale healthcare data.
- Built and deployed secure, interactive **healthcare web applications** with **FHIR** integration on **AWS**, enhancing scalability, reliability, and clinical interoperability.

Research Assistant — Dec. 2022 – Feb. 2024

- Cleaned and preprocessed extensive healthcare datasets, generating insightful visualizations to support data-driven research.
- Developed a **computer vision**-based object detection system integrated with automated data acquisition pipelines, enabling efficient data collection and analysis.
- Built and deployed interactive web platform for **3D modeling** and predictive analytics, enhancing visualization of object detection outcomes and improving user experience.

Regenstrief Institute Indianapolis, IN

Application Programmer Jun. 2025 – Aug. 2025

- Standardized U.S. EHR data by architecting pipelines for **OMOP** and **PCORnet** models, improving data quality and harmonization.
- Engineered automated, **scalable ETL pipelines** and extracted **longitudinal clinical features** from raw EHR data using **distributed cluster computing**.
- Created web-based **analytics dashboards** to support clinicians in interpreting complex data and improving decision-making.

Publications

1. Jiayi Tong, Lu Li, Dazheng Zhang, Kyra S. O'Brien, Yiwen Lu, Qiong Wu, Ting Zhou, Bingyu Zhang, Yu Huang, **Yu Hu**, et al. "FLAT-WHITE: from target trial emulation to federated high-throughput causal machine learning using decentralized real-world data." npj Health Syst. 3, 102 (2026).
2. Qiong Wu, Lu Li, Yuqing Lei, Ting Zhou, Huiling Tang, Bingyu Zhang, **Yu Hu**, et al. "Discovering repurposable drugs for Alzheimer's disease and related dementias: target trial emulation using decentralised real-world data." eBioMedicine of THE LANCET, Volume 132, 2026, 106466, ISSN 2352-3964.
3. Qinglin Gou*, **Yu Hu***, et al. "Adaptation of a fair individualized polysocial risk score for hospitalization risk prediction." JAMIA Open, Volume 9, Issue 4, August 2026, ooag161.
4. Yu Huang, Jie Xu, Zhengkang Fan, **Yu Hu**, et al. "Identifying Alzheimer's Disease Progression Subphenotypes via a Graph-based Framework using Electronic Health Records." J Healthc Inform Res 10, 317–339 (2026).
5. He X, Huang Y, **Yu Hu**, et al. "Toward Integrating Machine Learning-powered Polysocial Risk Scores into Electronic Health Record Workflows." AMIA Annu Symp Proc. 2025 May 22;2024:451-460. PMID: 41726481; PMCID: PMC12919456.
6. Zhou T, Chen A, **Hu Yu**, et al. "Automating Lung-RADS Categorization And Follow-Up Recommendations Using In-Context Learning With Large Language Models." AMIA Annu Symp Proc. 2025 May 22;2024:1567-1576. PMID: 41726431; PMCID: PMC12919574.
7. Jin Wang*, **Yu Hu***, et al. "Technical note: ShinyAnimalCV: open-source cloud-based web application for object detection, segmentation, and three-dimensional visualization of animals using computer vision." Journal of Animal Science, vol. 102, 2024, skad416.

*These authors contributed equally.

Projects

Clinical LLM Evaluation and Model Selection for Readmission Review

Apr. 2026 – Present

- Led a head-to-head evaluation of **Gemma 4 31B** and **MedGemma** for patient-level readmission review across **30K–80K-token** context windows, generating structured answers to clinical review questions with supporting rationales.
- Identified a key **clinical safety failure mode**: the domain-tuned model inferred unsupported answers from general medical knowledge (e.g., asserting prescriptions were sent to a pharmacy with no documentation), while Gemma 4 abstained with evidence-specific rationales, providing evidence for model selection.
- Validated outputs against **clinician review**: Gemma 4 corrected **7 of 10** answers the clinical team flagged as incorrect; analyzed the remaining disagreements to separate model errors from reviewer inference, and identified note-type confusion (outpatient vs. discharge instructions) as the next improvement target.
- Built an **LLM-as-a-Judge** framework using **Llama 3.3-70B** to grade **Gemma 4** outputs across four rubric dimensions with structured JSON verdicts, section-level scoring, and an interactive **HTML** report routing flagged patients to clinician review.
- Benchmarked long-context inference on **NVIDIA B200** GPUs (FP16 vs. FP8, tensor parallelism, memory settings); traced a throughput ceiling to vLLM's KV-cache handling of Gemma 4 **sliding-window attention** and reached **30–50 tokens/s** at 60K context with FP8.

Clinical LLMOps Platform for EHR Note Processing

Feb. 2026 – Present

- Generalized hard-coded, project-specific clinical NLP scripts into a reusable, **config-driven LLM platform** now shared by three clinical programs (**RN Health Coach**, **Bridge Clinic** readmission, **Transitional Care**); migrated all runtime settings into **YAML** with a `ConfigWrapper` supporting dotted-attribute and mapping access, so new cohorts onboard through configuration rather than code changes.
- Designed a modular **workflow architecture** (schema-driven data loading → prompt assembly → batched inference → post-processing → reporting) with a manager/processor per stage, checkpoint/resume, structured logging, and asynchronous concurrency control; deployed on the **HiPerGator** HPC cluster via **SLURM** and **vLLM**.
- Re-engineered the clinical note **LLM summarizer** into an **OpenAI-compatible API** service that runs on both self-hosted vLLM and the UF **NaviGator API**, with enforced **JSON Schema** output contracts, **Pydantic** validation, token-aware dynamic prompt budgeting, and automatic retry/fallback on malformed generations.
- Built a reusable base **HTML report generator** and pluggable LLM analysis sections, so programs extend shared components instead of maintaining separate reporting code.
- Established **MLOps** practices across projects (versioned shared libraries on internal **GitLab**, automated daily scheduling and logging, reproducible run artifacts), reducing new-pipeline setup from weeks to days.

GraphRAG Knowledge Graph for Heart Failure Readmission

May. 2026 – Present

- Built an end-to-end **GraphRAG** pipeline over a heart failure cohort of **637K clinical notes** from **3,946 patients**, including de-duplication of daily EHR snapshots, a three-layer note filter (note type / authoring provider type / length), and quota-based cohort sampling for documentation completeness.
- Performed **LLM-based entity and relation extraction** with **MedGemma-27B** served via **vLLM**, using a clinical prompt with a controlled entity-type and assertion (present/negated/historical) vocabulary; extracted **143K entities**, **71K relations**, and **243K attributes** from 5.5K text units.
- Constructed the patient knowledge graph with hierarchical **Leiden community detection** (2,040 communities across 7 levels) and map-reduce **LLM community reports** on readmission themes; hardened the pipeline against API rate-limit saturation, silent chunk drops, and PHI leakage into summaries.

Neonatal Lactation and Milk Intake Data Processing Project

Sep. 2025 – Present

- Standardized raw EHR lactation and neonatal milk intake data (pumping records, feeding documentation, inconsistent entries) and built automated feature engineering pipelines.
- Partnered with neonatologists and lactation specialists to define and validate **clinically meaningful features**, such as NPO patterns, respiratory support status, maternal milk availability, feeding intolerance, and early nutrition indicators.
- Developed machine learning models for early identification of infants at risk for inadequate milk intake or feeding complications; revalidated data and feature definitions after source updates, retrained the models, and authored a model brief.
- Deployed model and data pipelines on on-premise servers for reproducible, continuously updated analytics, and connected **Power BI** dashboards to model outputs for clinicians and lactation teams.

Graph-Based Modeling for Alzheimer's Disease Progression Prediction

Dec. 2023 – Nov. 2025

- Construct a patient similarity graph from **Electronic Health Records (EHR)** using **K-Nearest Neighbors (KNN)** to identify the top 100 most similar patient encounters.
- Utilize **NetworkX** to build the graph structure and integrate it with a **Graph Neural Network (GNN)** for predictive analysis of Alzheimer's Disease progression.
- Developed deep learning models using **PyTorch** to enhance predictive algorithms. Optimized model training and inference speed using **GPU CUDA** programming.
- Identify disease progression subphenotypes to improve early detection of Alzheimer's Disease and support personalized treatment strategies.

Advanced EHR Continuity Integration for Robust T2D Risk Prediction

Feb. 2024 – Nov. 2025

- Engineered an automated, **PCORnet**-based pipeline to systematically extract disease phenotypes, integrate longitudinal patient encounters, and execute comprehensive data cleaning on linked EHR and claims datasets from the **OneFlorida+ Clinical Research Network**.
- Developed and validated a machine learning framework for **type 2 diabetes** risk prediction, leveraging the **Harmonized Encounter Proportion Score (HEPS)** to quantify EHR data-continuity and improve predictive accuracy.
- Conducted fairness assessments using advanced algorithmic fairness metrics to identify disparities in model performance across racial-ethnic groups, ensuring equitable risk prediction.
- Utilized **cluster computing** and **parallel computation** techniques to optimize the system's computational efficiency, significantly reducing processing time for large-scale datasets.

OCT-Driven Retinal Image Biomarker Analysis for ADRD Prediction

Dec. 2023 – Sep. 2025

- Develop an automated pipeline to process **OCT DICOM scans**, enabling precise extraction and quantification of retinal imaging data.
- Construct predictive models using retinal thickness metrics from OCT scans to assess **ADRD onset risk**, establishing a predictive link between ocular biomarkers and neurodegenerative progression.
- Integrate OCT-derived features into **machine learning frameworks** to evaluate the predictive relationship between retinal measurements and ADRD development, enhancing early detection capabilities.

Healthcare Risk Prediction Web Application

Nov. 2023 – Sep. 2025

- Engineer front-end components using **React.js**, implementing secure user authentication and patient selection interfaces to enhance user experience.
- Develop a back-end **Flask** API for model integration, leveraging causal inference models to assess patient risks within the prediction framework.
- Utilize **SQL** to query and extract relevant data from healthcare databases, ensuring accurate and timely information retrieval.
- Integrate **FHIR-based database** and **EPIC system** inquiry functionality to streamline patient data retrieval and improve interoperability with electronic health records, while containerizing and orchestrating the entire software stack using **Docker**.

OMOP-FHIR Database Integration and Deployment Platform

Apr. 2024 – Sep. 2025

- Design and develop a comprehensive **OMOP database** to support standardized healthcare data integration and advanced analytics.
- Deploy and evaluate a web-based platform integrating **OMOP** and **FHIR** query capabilities, enhancing data interoperability and enabling real-time access.
- Utilize **ontology concept codes** to query, debug, and maintain the database, ensuring data consistency and accuracy.

Skills

LLM & NLP: Prompt Engineering, RAG/GraphRAG, LLM-as-a-Judge, vLLM, LLM Agents, HuggingFace Transformers, LangChain, spaCy, FAISS

Clinical Data Standards: UMLS, SNOMED CT, RxNorm, OMOP, FHIR, PCORnet

MLOps/LLMOps: SLURM, Model Serving & Monitoring, CI/CD, Structured Output (JSON Schema/Pydantic), Config-driven Pipelines

Programming Languages: Python, SQL, R, C/C++, Java, JavaScript, HTML, CSS, LaTeX, Shell, Matlab

Machine Learning & Data Analysis: PyTorch, TensorFlow, Scikit-learn, XGBoost, NetworkX, Causal Inference

Tools & Platforms: Docker, Linux, Tableau, Power BI, Cluster Computing, Cloud Computing, Git

Awards and Honors

- **Honor**: Herbert Wertheim College of Engineering Achievement Award Scholarship (Awarded for academic excellence and research contributions) — 2021
- **Scholarship**: Undergraduate Scholarship awarded to top 5% of students for academic excellence in Electrical Engineering. — 2020

Academic Services

Reviewers for: *The Fourteenth IEEE International Conference on Healthcare Informatics*

Reviewers for: *IEEE International Conference on Healthcare Informatics - ICHI 2025*

Reviewers for: *The 12th IEEE International Conference on Healthcare Informatics*